

Week 3

Date- 20 to 25 july

Day 1:-

Objective

Understand modern JavaScript features that simplify development and improve code quality.

Topics Covered

- **let and const**
- **Arrow Functions**
- **Template Literals**
- **Destructuring Assignment**
- **Spread and Rest Operators**
- **Default Parameters**
- **Modules**

Practical Work

Developed utility programs using ES6 features:

- **Student Record Manager**
- **Employee Salary Calculator**
- **Dynamic Object Manipulation**

Day 2:-

Objective

Understand asynchronous programming concepts used in modern web applications.

Topics Covered

- **Callbacks**
- **Promises**
- **Async & Await**
- **Fetch API**
- **JSON**
- **Error Handling**

Practical Work

Created:

- **Weather Information Application**
- **Random User Profile Viewer**

Day 3:-

Objective

Understand the basics of React and component-based development.

Topics Covered

- **Introduction to React**
- **Virtual DOM**
- **JSX**
- **Components**
- **Project Structure**
- **npm Packages**

Practical Work

- **Installed React using Vite**
- **Created first React application**

Day 4:-

Objective

Learn how to manage data and user interactions in React.

Topics Covered

- **Functional Components**
- **Props**
- **useState Hook**
- **Event Handling**
- **Conditional Rendering**
- **Lists & Keys**

Practical Work

Developed:

- **Counter Application**
- **Student Information Dashboard**

Day 5:-

Objective

Build interactive forms and navigate between multiple pages.

Topics Covered

- **Controlled Components**
- **Form Validation**
- **React Router**
- **Navigation**
- **Dynamic Routes**

Practical Work

Developed:

- **Login Page**
- **Registration Form**

Day 6,7:-

Objective

Display real-time data inside React applications using REST APIs.

Topics Covered

- **REST APIs**
- **Loading States**
- **Error Handling**
- **API Integration**

Objective

Revise all concepts learned during the week and apply them in a complete React project.